

Discussion-3 manipulate data

Discussion Topic:

One of the most important functions of any relational database management system (RDBMS) is to support your ability to manipulate data.

To this end, you must be able to insert data in a table, modify that data, and then delete whatever data you no longer want to store.

What are some of the MySQL commands that can be used to perform these tasks?

Give an example of each command using an SQL statement and explain how each command works.

What are some of the hazards of using MySQL commands to update and delete data?

My Post:

Hello Class,

One of the core functions of a Relational Database Management System (RDBMS) is to allow users to manipulate data, such as inserting data into a table, modifying that data, and then deleting whatever data they no longer want to store. These RDBMS functions are handled by the Data Manipulation Language (DML), a subset of the Structured Query Language (SQL), and implementations of SQL like MySQL (Murach, 2019). Note that the Data Definition Language (DDL) is also a subset of SQL, which handles the definition or modification of the structure of database objects (schemas) like tables and indexes. In other words, DML is used to manipulate the data within schemas, and DDL is used to manipulate the schemas. This article focuses on the DML operations (INSERT, UPDATE, and DELETE) used to manipulate data within MySQL tables, providing examples.

DML Operations

The DML operations for modifying data in MySQL tables are handled by the statements (also referred to as commands): `INSERT`, `UPDATE`, and `DELETE`.

- The `INSERT` command adds new rows of data.
- The `UPDATE` command modifies data in existing rows.
- The `DELETE` command removes rows entirely.

Note that these commands manipulate rows, not columns. However, the command `SELECT` is used for querying and retrieving data from a table based on the table column (Watt & Eng, 2014).

These commands (`INSERT`, `UPDATE`, `DELETE`) are often used in combination with the `SELECT` command and the `WHERE` clause to manipulate data by querying the table columns and selecting rows based on a specific condition(s). Note that the `WHERE` clause is responsible for filtering rows, also called records, based on specific conditions (Kathuria, 2021).

Data Manipulation Examples

To illustrate the examples on how to use the `INSERT`, `UPDATE`, and `DELETE` commands, the following `products` table is going to be used.

Figure 1

Products Table

product_id	category_id	product_code	product_name	description	list_price	discount_percent	date_added
1	1	strat	Fender Stratocaster	The Fender Stratocaster is the electric guitar de...	699.00	30.00	2017-10-30 09:32:40
2	1	les_paul	Gibson Les Paul	This Les Paul guitar offers a carved top and hu...	1199.00	30.00	2017-12-05 16:33:13
3	1	sg	Gibson SG	This Gibson SG electric guitar takes the best of t...	2517.00	52.00	2018-02-04 11:04:31
4	1	fg700s	Yamaha FG700S	The Yamaha FG700S solid top acoustic guitar ha...	489.99	38.00	2018-06-01 11:12:59
5	1	washburn	Washburn D10S	The Washburn D10S acoustic guitar is superbly ...	299.00	0.00	2018-07-30 13:58:35
6	1	rodriguez	Rodriguez Caballero 11	Featuring a carefully chosen, solid Canadian ce...	415.00	39.00	2018-07-30 14:12:41
7	2	precision	Fender Precision	The Fender Precision bass guitar delivers the so...	799.99	30.00	2018-06-01 11:29:35
8	2	hofner	Hofner Icon	With authentic details inspired by the original, t...	499.99	25.00	2018-07-30 14:18:33
9	3	ludwig	Ludwig 5-piece Drum Set with Cymbals	This product includes a Ludwig 5-piece drum set...	699.99	30.00	2018-07-30 12:46:40
10	3	tama	Tama 5-Piece Drum Set with Cymbals	The Tama 5-piece Drum Set is the most afforda...	799.99	15.00	2018-07-30 13:14:15

Note: The figure table illustrates a product table of various types of guitars.

The INSERT Commands

The `INSERT` command is used to insert rows/records (to store new data) in tables. Different syntax variations of the `INSERT` command exist in MySQL:

Using VALUES

```
INSERT INTO table_name
(column1, column2, ...) -- the columns list
VALUES
(value1a, value2a, ...), -- Values for the first row
(value1b, value2b, ...), -- Values for the second row
(value1c, value2c, ...), -- Values for the third row
...
(value1n, value2n, ...); -- Values for the last row
```

Using SET

```
INSERT INTO table_name
SET column1 = value1,
    column2 = value2,
    ... ;
```

With the `SET` command, only one value can be inserted by column.

Using SELECT

```
INSERT INTO table_name (column1, column2,...)
SELECT source_column1, source_column2,...
FROM source_table
WHERE condition;
```

Here, `SELECT` is used to insert columns from a source table column where a condition is met.

INSERT Example using the products table:

Figure 2

INSERT Example

```
1
2 • INSERT INTO products
3   SET
4     category_id = 1,
5     product_code = 'tele',
6     product_name = 'Fender Telecaster',
7     description = 'Iconic electric guitar known for its bright, cutting tone.',
8     list_price = 899.00,
9     discount_percent = 15.00,
10    date_added = NOW();
11
12
13 • SELECT *
14   From products;
15
```

product_id	category_id	product_code	product_name	description	list_price	discount_percent	date_added
1	1	strat	Fender Stratocaster	The Fender Stratocaster is the electric guitar de...	699.00	30.00	2017-10-30 09:32:40
2	1	les_paul	Gibson Les Paul	This Les Paul guitar offers a carved top and hu...	1199.00	30.00	2017-12-05 16:33:13
3	1	sg	Gibson SG	This Gibson SG electric guitar takes the best of t...	2517.00	52.00	2018-02-04 11:04:31
4	1	fg700s	Yamaha FG700S	The Yamaha FG700S solid top acoustic guitar ha...	489.99	38.00	2018-06-01 11:12:59
5	1	washburn	Washburn D10S	The Washburn D10S acoustic guitar is superbly ...	299.00	0.00	2018-07-30 13:58:35
6	1	rodriguez	Rodriguez Caballero 11	Featuring a carefully chosen, solid Canadian ce...	415.00	39.00	2018-07-30 14:12:41
7	2	precision	Fender Precision	The Fender Precision bass guitar delivers the so...	799.99	30.00	2018-06-01 11:29:35
8	2	hofner	Hofner Icon	With authentic details inspired by the original, t...	499.99	25.00	2018-07-30 14:18:33
9	3	ludwig	Ludwig 5-piece Drum Set with Cymbals	This product includes a Ludwig 5-piece drum set...	699.99	30.00	2018-07-30 12:46:40
10	3	tama	Tama 5-Piece Drum Set with Cymbals	The Tama 5-piece Drum Set is the most afforda...	799.99	15.00	2018-07-30 13:14:15
11	1	tele	Fender Telecaster	Iconic electric guitar known for its bright, cuttin...	899.00	15.00	2025-04-28 21:27:23

Note: The figure illustrates the SQL code for inserting a new guitar item in the products table.

The UPDATE Commands

The `UPDATE` command is used to update existing rows/records' values (to modify existing data) in tables.

Syntax:

```
UPDATE table_name
SET column1 = new_value1,
    column2 = new_value2
WHERE 'condition'; -- The condition determines which rows are updated
```

The `WHERE` clause is used to update specific records' values based on a specific condition.

UPDATE Example using the products table:

Figure 3

UPDATE Example

```
1
2 * UPDATE products
3   SET discount_percent = discount_percent + 10
4   WHERE category_id = 1;
5
6
7 * SELECT *
8   From products;
9
```

	product_id	category_id	product_code	product_name	description	list_price	discount_percent	date_added
▶	1	1	strat	Fender Stratocaster	The Fender Stratocaster is the electric guitar de...	699.00	40.00	2017-10-30 09:32:40
	2	1	les_paul	Gibson Les Paul	This Les Paul guitar offers a carved top and hu...	1199.00	40.00	2017-12-05 16:33:13
	3	1	sg	Gibson SG	This Gibson SG electric guitar takes the best of t...	2517.00	62.00	2018-02-04 11:04:31
	4	1	fg700s	Yamaha FG700S	The Yamaha FG700S solid top acoustic guitar ha...	489.99	48.00	2018-06-01 11:12:59
	5	1	washburn	Washburn D10S	The Washburn D10S acoustic guitar is superbly ...	299.00	10.00	2018-07-30 13:58:35
	6	1	rodriguez	Rodriguez Caballero 11	Featuring a carefully chosen, solid Canadian ce...	415.00	49.00	2018-07-30 14:12:41
	7	2	precision	Fender Precision	The Fender Precision bass guitar delivers the so...	799.99	30.00	2018-06-01 11:29:35
	8	2	hofner	Hofner Icon	With authentic details inspired by the original, t...	499.99	25.00	2018-07-30 14:18:33
	9	3	ludwig	Ludwig 5-piece Drum Set with Cymbals	This product includes a Ludwig 5-piece drum set...	699.99	30.00	2018-07-30 12:46:40
	10	3	tama	Tama 5-Piece Drum Set with Cymbals	The Tama 5-piece Drum Set is the most afforda...	799.99	15.00	2018-07-30 13:14:15
	11	1	tele	Fender Telecaster	Iconic electric guitar known for its bright, cuttin...	899.00	25.00	2025-04-28 21:27:23

Note: The figure illustrates the SQL code for adding +10 to the discount_percent for all items with a category_id = 1

The DELETE Commands

The DELETE command is used to delete rows/records (to delete existing data) in tables. Syntax:

```
DELETE FROM table_name
WHERE 'condition'; -- Specifies row(s) to delete
```

DELETE Example using the products table:

Figure 3

DELETE Example

```
1
2 * DELETE FROM products
3   WHERE category_id = 1;
4
5 * SELECT *
6   From products;
7
```

	product_id	category_id	product_code	product_name	description	list_price	discount_percent	date_added
	7	2	precision	Fender Precision	The Fender Precision bass guitar delivers the so...	799.99	30.00	2018-06-01 11:29:35
	8	2	hofner	Hofner Icon	With authentic details inspired by the original, t...	499.99	25.00	2018-07-30 14:18:33
	9	3	ludwig	Ludwig 5-piece Drum Set with Cymbals	This product includes a Ludwig 5-piece drum set...	699.99	30.00	2018-07-30 12:46:40
	10	3	tama	Tama 5-Piece Drum Set with Cymbals	The Tama 5-piece Drum Set is the most afforda...	799.99	15.00	2018-07-30 13:14:15

Note: The figure illustrates the SQL code for deleting all items with a category_id = 1

Hazards of Using UPDATE and DELETE

When using the `UPDATE` and `DELETE` commands, users need to be extremely careful, as these commands are very powerful and consequently come with significant risks. Accidentally misusing these commands can result in widespread data corruption or data loss. Some of these hazards include

- Missing or incorrect `WHERE` clause resulting in the `UPDATE` or `DELETE` commands being intentionally applied to all rows.
- Data loss caused by incorrect logic or use of the `WHERE` clause condition and the `UPDATE` command, which may be potentially fixed, and the `DELETE` command, which often results in irreversible loss of data.

To summarize, the `INSERT`, `UPDATE`, and `DELETE` commands are DML operations in MySQL for adding, modifying, and removing data within tables. The `INSERT` syntax can take various forms. The `UPDATE` and `DELETE` commands require particular attention due to their potential to modify or destroy existing data on a large scale.

-Alex

References:

Kathuria, H. (2021, November 9). How to write a `WHERE` clause in SQL. Learn SQL.
<https://learnsql.com/blog/where-clause-in-sql/>

Murach, J. (2019). Chapter 1: An introduction to relational databases. *Murach's MySQL (3rd ed.)*. Murach Books. ISBN: 9781943872367

Watt, A., & Eng, N. (2014). Chapter 16 SQL Data Manipulation Language. Database Design – 2nd Edition. Open textbooks, British Columbia University Open Campus.
<https://opentextbc.ca/dbdesign01/chapter/chapter-sql-dml/>